

Immune system disorders:

The incidence of acute graft versus host disease (a-GVHD) data was collected in allogeneic patients (n=28). A total of 14 patients (50%) experienced a-GVHD. The incidence of a-GVHD grades I-II was 46.4% (13/28), while the incidence of grade III-IV was 3.6% (1/28). Chronic GVHD was reported only if it is the cause of death: one patient died 13 months post-transplant.

Infections and infestations:

Infections (documented and non-documented febrile neutropenia) were experienced in 89% of patients (49/55). Mild/moderate fever was reported in 76% of patients.

Hepato-biliary disorders:

Grade 3 elevated transaminases were reported in 24% of patients.

Veno occlusive disease (VOD) was reported in 15% (4/27) and 7% (2/28) of the autologous and allogenic transplant respectively. VOD observed were neither fatal nor severe and resolved in all cases.

Busulfan Injection in combination with fludarabine (FB)

In Adults

The safety profile of Busulfan Injection combined with fludarabine (FB) has been examined through a review of adverse events reported in published data from clinical trials in RIC regimen. In these studies, a total of 1574 patients received FB as a reduced intensity conditioning (RIC) regimen prior to haematopoietic progenitor cell transplantation.

Myelo-suppression and immuno-suppression were the desired therapeutic effects of the conditioning regimen and consequently were not considered undesirable effects.

Infections and infestations:

The occurrence of infectious episodes or reactivation of opportunistic infectious agents mainly reflects the immune status of the patient receiving a conditioning regimen.

The most frequent infectious adverse reactions were Cytomegalovirus (CMV) reactivation [range: 30.7% - 80.0%], Epstein-Barr Virus (EBV) reactivation [range: 2.3% - 61%], bacterial infections [range: 32.0% - 38.9%] and viral infections [range: 1.3% - 17.2%].

Gastrointestinal disorders:

The highest frequency of nausea and vomiting was 59.1% and the highest frequency of stomatitis was 11%.

Renal and urinary disorders:

It has been suggested that conditioning regimens containing fludarabine were associated with higher incidence of opportunistic infections after transplantation because of the immunosuppressive effect of fludarabine. Late haemorrhagic cystitis occurring 2 weeks post-transplant are likely related to viral infection / reactivation. Haemorrhagic cystitis including haemorrhagic cystitis induced by viral infection was reported in a range between 16% and 18.1%.

Hepato-biliary disorders:

VOD was reported with a range between 3.9% and 15.4%.

The treatment-related mortality/non-relapse mortality (TRM/NRM) reported until day+100 post-transplant has also been examined through a review of published data from clinical trials. It was considered as deaths that could be attributable to secondary side effects after HPCT and not related to the relapse/progression of the underlying haematological malignancies.

The most frequent causes of reported TRM/NRMs were infection/sepsis, GVHD, pulmonary disorders and organ failure.

Tabulated summaries of adverse reactions

Frequencies are defined as: very common (≥ 1/10), common (≥ 1/10, < 1/10), uncommon (≥ 1/1,000, < 1/100) or not known (cannot be estimated from the available data). Undesirable effects coming from post-marketing survey have been implemented in the tables with the incidence "not known".

Busulfan Injection in combination with cyclophosphamide or melphalan

Adverse reactions reported both in adults and paediatric patients as more than an isolated case are listed below, by system organ class and by frequency. Within each frequency grouping, adverse events are presented in order of decreasing seriousness.

System organ class	Very common	Common	Uncommon	Not known
Infections and infestations	Rhinitis Pharyngitis			
Blood and lymphatic system disorders	Neutropenia Thrombocytopenia Febrile neutropenia Anaemia Pancytopenia			
Immune system disorders	Allergic reaction			
Endocrine disorders				Hypogonadism **
Metabolism and nutrition disorders	Anorexia Hyperglycaemia Hypocalcaemia Hypokalaemia Hypomagnesaemia Hypophosphatemia	Hyponatraemia		
Psychiatric disorders	Anxiety Depression Insomnia	Confusion	Delirium Nervousness Hallucination Agitation	
Nervous system disorders	Headache Dizziness		Seizure Encephalopathy Cerebral haemorrhage	
Eye disorders				Cataract Corneal thinning Lens disorders ***
Cardiac disorders	Tachycardia	Arrhythmia Atrial fibrillation Cardiomegaly Pericardial effusion Pericarditis	Ventricular extrasystoles Bradycardia	
Vascular disorders	Hypertension Hypotension Thrombosis Vasodilatation		Femoral artery thrombosis Capillary leak syndrome	
Respiratory thoracic and mediastinal disorders	Dyspnoea Epistaxis Cough Hiccup	Hyperventilation Respiratory failure Alveolar haemorrhages Asthma Atelectasis Pleural effusion	Hypoxia	Interstitial lung disease**
Gastrointestinal disorders	Stomatitis Diarrhoea Abdominal pain Nausea Vomiting Dyspepsia Ascites Constipation Anus discomfort	Haematemesis Ileus Oesophagitis	Gastrointestinal haemorrhage	Tooth hypoplasia**
Hepato-biliary disorders	Hepatomegaly Jaundice	Veno occlusive liver disease *		
Skin and subcutaneous tissue disorders	Rash Pruritis Alopecia	Skin desquamation Erythema Pigmentation disorder		
Musculoskeletal and connective tissue disorders	Myalgia Back pain Arthralgia			
Renal and urinary disorders	Dysuria Oligurea	Haematuria Moderate renal Insufficiency		
Reproductive system and breast disorders				Premature menopause Ovarian failure**

General disorders and administration site conditions	Asthenia Chills Fever Chest pain Oedema Oedema general Pain Pain or inflammation at injection site Mucositis			
Investigations	Transaminases increased Bilirubin increased GGT increased Alkaline phosphatases increased Weight increased Abnormal breath sounds Creatinine elevated	Bun increase Decrease ejection fraction		

* veno occlusive liver disease is more frequent in paediatric population.

** reported in post marketing with IV busulfan

*** reported in post marketing with oral busulfan

Busulfan Injection in combination with fludarabine (FB)

The incidence of each adverse reactions presented in the following table has been defined according to the highest incidence observed in published clinical trials in RIC regimen for which the population treated with FB was clearly identified, whatever the schedules of busulfan administrations and endpoints. Adverse reactions reported as more than an isolated case are listed below, by system organ class and by frequency.

System organ class	Very common	Common	Not known*
Infections and infestations	Viral infection CMV reactivation EBV reactivation Bacterial infection	Invasive fungal infection Pulmonary infection	Brain abscess Cellulitis Sepsis
Blood and lymphatic system disorders			Febrile neutropenia
Metabolism and nutrition disorders	Hypoalbuminaemia Electrolyte disturbance Hyperglycaemia		Anorexia
Psychiatric disorders			Agitation Confusional state Hallucination
Nervous system disorders		Headache Nervous system disorders [Not Elsewhere Classified]	Cerebral haemorrhage Encephalopathy
Cardiac disorders			Atrial fibrillation
Vascular disorders		Hyper-tension	
Respiratory thoracic and mediastinal disorders		Pulmonary haemorrhage	Respiratory failure
Gastro-intestinal disorders	Nausea Vomiting Diarrhoea Stomatitis		Gastro-intestinal haemorrhage Tooth hypoplasia*
Hepato-biliary disorders	Veno occlusive liver disease		Jaundice Liver disorders
Skin and subcutaneous tissue disorders		Rash	
Renal and urinary disorders	Haemorrhagic cystitis**	Renal disorder	Oliguria
General disorders and administration site conditions	Mucositis		Asthenia Oedema Pain
Investigations	Transaminases increased Bilirubine increased Alkaline phosphatases increased	Creatinine elevated	Blood lactate dehydrogenase increased Blood uric acid increased Blood urea increased GGT increased Weight increased

* reported in post marketing experience

** include haemorrhagic cystitis induced by viral infection

Overdose and Treatment

There is no known antidote to Busulfan other than hematopoietic progenitor cell transplantation. In the absence of hematopoietic progenitor cell transplantation, the recommended dosage for Busulfan would constitute an overdose of busulfan. The principal toxic effect is profound bone marrow hypoplasia/aplasia and pancytopenia, but the central nervous system, liver, lungs, and gastrointestinal tract may be affected. Monitor hematologic status closely and institute vigorous supportive measures as medically indicated.

Survival after a single 140 mg dose of Tablets in an 18 kg, 4-year old child has been reported. Inadvertent administration of a greater than normal dose of oral busulfan (2.1 mg/kg; total dose of 23.3 mg/kg) occurred in a 2-year old child prior to a scheduled bone marrow transplant without sequelae. An acute dose of 2.4 g was fatal in a 10-year old boy. There is one report that busulfan is dialyzable, thus dialysis should be considered in the case of overdose.

Effects on Ability to Drive and Use Machine

Not relevant

Incompatibilities

Busulfan must be diluted prior to use with either 0.9% Sodium Chloride Injection, (normal saline) or 5% Dextrose Injection, Busulfan should not be mixed with other medicinal products.

Storage Conditions

Unopened vials of Busulfan Injection must be stored under refrigerated conditions between 2°C to 8°C (36°F to 46°F). Discard unused portion.

Diluted solution:

8 hours (including infusion time) after dilution when stored at 20 °C ± 5 °C.

12 hours after dilution when stored at 2 °C-8 °C followed by 3 hours stored at 20 °C ± 5 °C (including infusion time).

From a microbiological point of view, the product should be used immediately after dilution. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than the above-mentioned conditions when dilution has taken place in controlled and validated aseptic conditions.

Shelf life

2 years

Packaging Available

Each carton contains 1 vial (10 ml/ 20 mm neck flint clear tubular USP type- I glass vial with reinforced Non-PVC Base, stoppered with 20 mm rubber stopper and sealed with 20 mm Aluminum Flip off seal).

MAL22116020AZ

Name and Address of The Manufacturer

Shilpa Medicare Limited,
Plot No. S-20 to S-26, Pharma SEZ TSIIIC,
Green Industrial Park, Polepally Village,
Jadcherla Mandal, Mahabubnagar District,
Telangana State, India. Pin 509301.

Name & Address of Product Registration Holder (PRH):

Unimed Sdn Bhd
53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara 52200,
Kuala Lumpur, Malaysia.

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